Guided Problem Solving - 7 & 8

Foundations of Fluid Mechanics, AM2530

October 31, 2026

Everything you need is in this paper: the formulae, loss coefficients, roughness values and water properties are on page 5, and the Moody chart is on page 6. Take $g = 9.81\,\mathrm{m/s^2}$.

1. You have bought a house and want to improve the flow of water to the top floor. The flow is poor for three reasons: the city water pressure at the water meter is only 200 kPa (gauge); the existing piping has a small diameter ($D = 12.7\,\mathrm{mm}$) and has crudded up, raising its relative roughness to $e/D = 0.05$; and the top-floor tap is 15 m above the meter. From the meter the pipe runs 7 m horizontally, rises 15 m and runs 1 m horizontally to the tap, which discharges to atmosphere (Figure 1). Two improvements are proposed. In Option 1, all the piping after the meter is replaced by new smooth pipe ($e/D = 0$) of diameter 19 mm. In Option 2, the old piping is kept and a booster pump at the foot of the riser delivers 300 kPa (gauge) at its outlet. The water is at 20°C. Neglect minor losses, and note that the velocity in the pipe at the meter (or at the pump outlet) equals the velocity at the tap. **Read all friction factors from the Moody chart.**

   (a) Find the flow rate (L/min) delivered to the top floor by the existing piping.

   (b) Find the flow rates for Option 1 and Option 2. Which option is more effective?

   (c) Mark the three operating points on the Moody chart. Use them to explain why the booster pump helps so little.

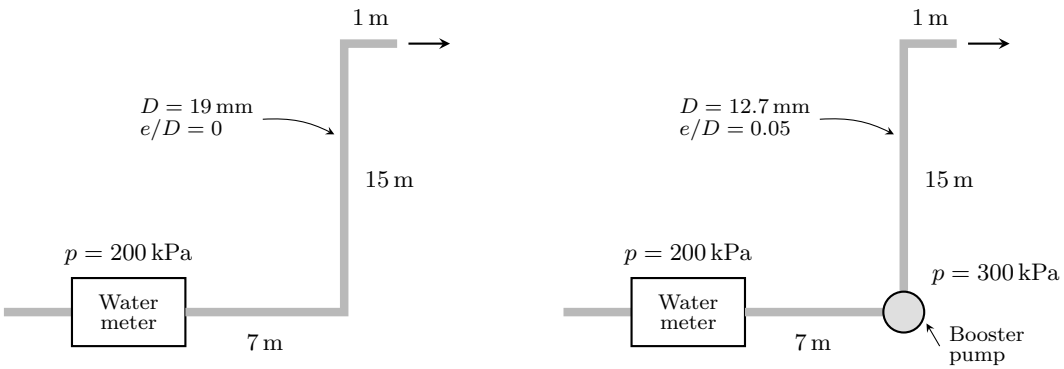


Figure 1: The two proposals. The existing piping follows the route of Option 1 but has $D = 12.7\,\mathrm{mm}$ and $e/D = 0.05$, with no pump.

2. Galvanized-iron drainpipes of diameter 50 mm are located at the four corners of a building, but three of them become clogged with debris (Figure 2). The roof area is $500\,\mathrm{m^2}$ and the building is 5 m high; the drainpipes run the full height of the building and both ends are open to atmosphere. Treat the working pipe as running full, with atmospheric pressure and the same velocity at its top and bottom, and ignore minor losses. The water is at 20°C. **Use the Colebrook formula for the friction factor.**

   (a) Find the rainfall rate (mm/h) at which the single working drainpipe can no longer drain the roof.

   (b) Show that, within these assumptions, the answer does not depend on the height of the building, and explain why physically.

(c) Find the capacity of a 75 mm drainpipe of the same material. Why does the capacity rise by more than the ratio of the pipe areas?

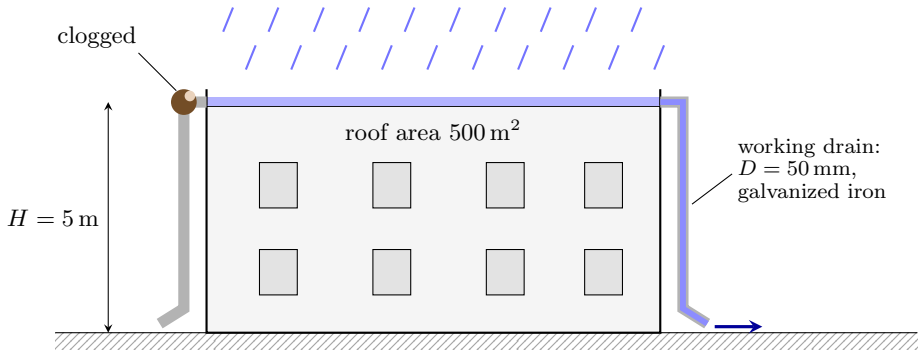


Figure 2: Rain draining from the roof. The two drains at the rear corners (not shown) are also clogged.

3. A new industrial plant requires a water flow rate of 5.7 m^3/min. The gauge pressure in the water main, located in the street 50 m from the plant, is 800 kPa, and the gauge pressure required in the plant is 500 kPa. The galvanized-iron supply line will have a total length of 65 m and four standard 90° elbows (Figure 3). The water is at 15°C. Neglect elevation changes and the loss at the junction with the main. **Use the Haaland formula for the friction factor.**

   (a) Find the minimum inside diameter of the supply line.

   (b) Galvanized pipe is available in nominal sizes of 4, 5 and 6 in, with inside diameters of 102.3, 128.2 and 154.1 mm. Which size should be installed? For this size, find the pressure delivered to the plant and the mean velocity in the line.

   (c) Would you have chosen the same size had the elbows been ignored? Use the scaling $\Delta p \propto fLQ^2/D^5$ to explain why one step in standard size changes the pressure drop so much.

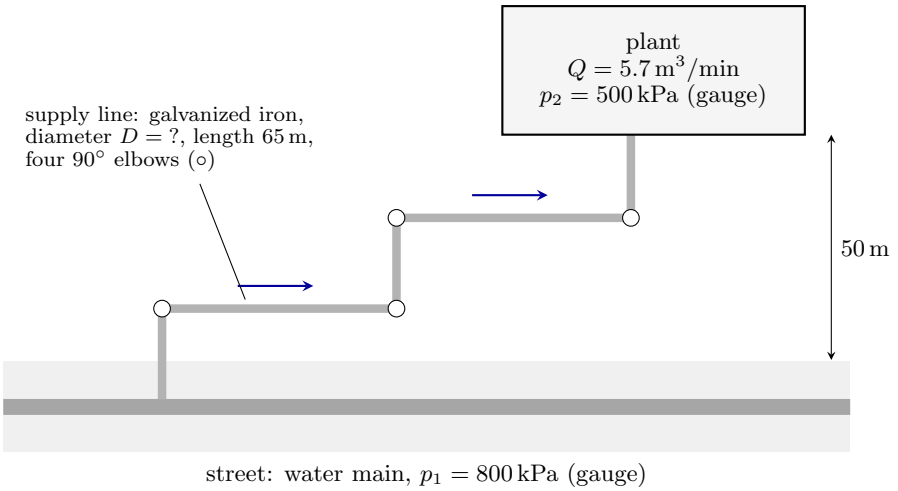


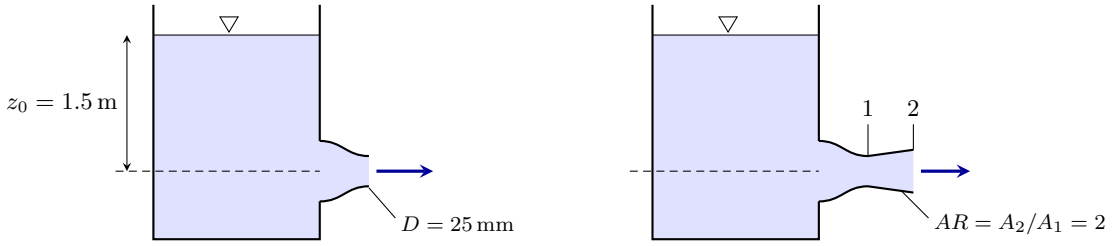
Figure 3: Plan view of the supply line from the main to the plant (not to scale).

4. Water rights granted to each citizen by the Emperor of Rome allowed a calibrated, circular bronze nozzle to be attached to the public water main. Some citizens fitted diffusers to the nozzle outlets to increase their discharge (Figure 4). The static head available from the main is $z_0 = 1.5$ m above the nozzle, the nozzle exit diameter is 25 mm, and the water discharges to atmosphere. The well-rounded nozzle entrance has $K = 0.04$. The diffuser has an exit-to-inlet area ratio $AR = 2.0$ and a loss coefficient $K_d = 0.30$; both loss coefficients are based on the velocity at the nozzle exit. Neglect pipe friction.

(a) Find the discharge (L/s) with the nozzle alone.

(b) Find the discharge with the diffuser attached, and the percentage increase.

(c) Find the gauge pressure at the nozzle exit (the diffuser inlet) when the diffuser is fitted. Use the result to explain how the diffuser raises the flow.



(a) Nozzle alone (b) Nozzle with diffuser

Figure 4: Roman nozzle on the water main, without and with a diffuser (not to scale).

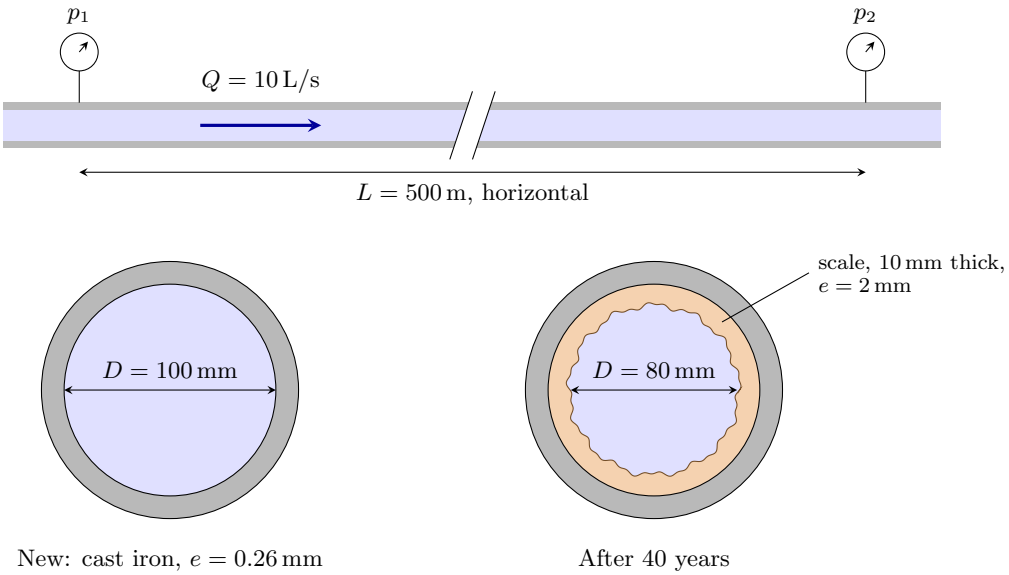
5. In hard-water districts, scale builds up on the inside of water mains over the years (Figure 5). A horizontal cast-iron main, 500 m long, had an inside diameter of 100 mm and a roughness $e = 0.26$ mm when new, and it carried its design flow of 10 L/s of water at 20°C. After 40 years of service a layer of scale about 10 mm thick lines the wall: the bore is now 80 mm, and the scale surface has an effective roughness of 2 mm. Neglect minor losses. **Use the Moody chart or the Haaland formula for the friction factor.**

   (a) Find the pressure drop along the main at the design flow when it was new.

   (b) Find the pressure drop at the same flow after 40 years. Which diameter enters each of V, Re, e/D and L/D in this calculation?

   (c) How much of the increase comes from the smaller bore, and how much from the rougher wall? Which matters more?

   (d) If the pressure drop available is still that of part (a), what flow rate does the old main deliver?



New: cast iron, $e = 0.26$ mm After 40 years

Figure 5: The main (top) and its cross-section when new and after 40 years of scale formation (not to scale).

6. Water at 20°C flows steadily from a large open tank through a length of smooth plastic tubing and discharges to atmosphere (Figure 6). The tubing leaves the base of the tank horizontally; its inside

diameter is 3.18 mm and its length is 15.3 m. For laminar flow take the kinetic energy coefficient $\alpha = 2$ and the entrance loss coefficient $K_{\text{ent}} = 1.4$.

(a) Find the largest volume flow rate (mL/s) for which the flow in the tubing remains laminar.

(b) Find the water depth in the tank below which the flow will be laminar.

(c) What fraction of this depth is spent overcoming friction in the tubing? Mark the operating point on the Moody chart.

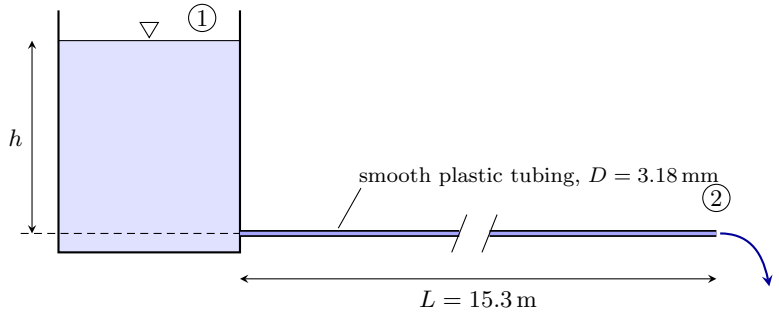


Figure 6: Laminar flow from a large tank through thin tubing (not to scale).

Data sheet

Energy equation between sections 1 and 2 of a pipe (steady, incompressible flow):

$$\left(\frac{p_1}{\rho} + \alpha_1 \frac{V_1^2}{2} + g z_1\right) - \left(\frac{p_2}{\rho} + \alpha_2 \frac{V_2^2}{2} + g z_2\right) = h_{lT} = h_l + h_{lm},$$

with $\alpha \approx 1$ for turbulent flow and $\alpha = 2$ for fully developed laminar flow. V is the mean velocity, $Q = VA$ and $Re = VD/\nu$.

Major loss (Darcy–Weisbach): $h_l = f \dfrac{L}{D} \dfrac{V^2}{2}$.

Friction factor

- Laminar flow ($Re < 2300$): $f = 64/Re$.
- Turbulent flow, Colebrook formula (implicit in f, so iterate):

$$\frac{1}{\sqrt{f}} = -2.0 \log_{10}\left(\frac{e/D}{3.7} + \frac{2.51}{Re\sqrt{f}}\right).$$

- Turbulent flow, Haaland formula (explicit; within about 2% of Colebrook):

$$\frac{1}{\sqrt{f}} = -1.8 \log_{10}\left[\left(\frac{e/D}{3.7}\right)^{1.11} + \frac{6.9}{Re}\right].$$

- Complete turbulence (fully rough limit, f independent of Re): $f = \left[-2.0 \log_{10} \dfrac{e/D}{3.7}\right]^{-2}$. Moody chart: page 6.

Minor losses: $h_{lm} = K \dfrac{V^2}{2}$ or $h_{lm} = f \dfrac{L_e}{D} \dfrac{V^2}{2}$, with V the mean velocity in the pipe (for a sudden area change, the larger of the two velocities).

Loss coefficients, K

Fitting	K
Entrance, re-entrant	0.78
Entrance, square-edged	0.50
Entrance, rounded, $r/D = 0.02$	0.28
Entrance, rounded, $r/D = 0.06$	0.15
Entrance, rounded, $r/D \geq 0.15$	0.04
Exit into a large reservoir	1.0
Sudden expansion (based on V_1)	$(1 - A_1/A_2)^2$
Sudden contraction (based on V_2):	
$A_2/A_1 = 0$	0.50
$A_2/A_1 = 0.2$	0.42
$A_2/A_1 = 0.4$	0.28
$A_2/A_1 = 0.6$	0.15
$A_2/A_1 = 0.8$	0.04

Equivalent lengths, L_e/D (fully open)

Fitting	L_e/D
Gate valve	8
Globe valve	340
Angle valve	150
Ball valve	3
Standard elbow, 90°	30
Standard elbow, 45°	16
Return bend, close pattern	50
Standard tee, through run	20
Standard tee, through branch	60

Roughness, e

Pipe material	e (mm)
Riveted steel	0.9–9
Concrete	0.3–3
Wood stave	0.2–0.9
Cast iron	0.26
Galvanized iron	0.15
Asphalted cast iron	0.12
Commercial steel	0.046
Drawn tubing	0.0015
Smooth plastic	≈ 0

Properties of water

T (°C)	ρ (kg/m^3)	μ (Pa s)	ν (m^2/s)	p_v (kPa)
10	1000	1.30×10^{-3}	1.30×10^{-6}	1.23
15	999	1.14×10^{-3}	1.14×10^{-6}	1.71
20	998	1.01×10^{-3}	1.01×10^{-6}	2.34

Unit conversions: $1\,\text{m}^3/\text{s} = 1000\,\text{L/s} = 6 \times 10^4\,\text{L/min}$;
$1\,\text{mm/h} = 2.78 \times 10^{-7}\,\text{m/s}$.

Data from Fox, McDonald and Pritchard, *Introduction to Fluid Mechanics*, 8th ed., Tables 8.1, 8.2, 8.4 and A.8 and Fig. 8.15.

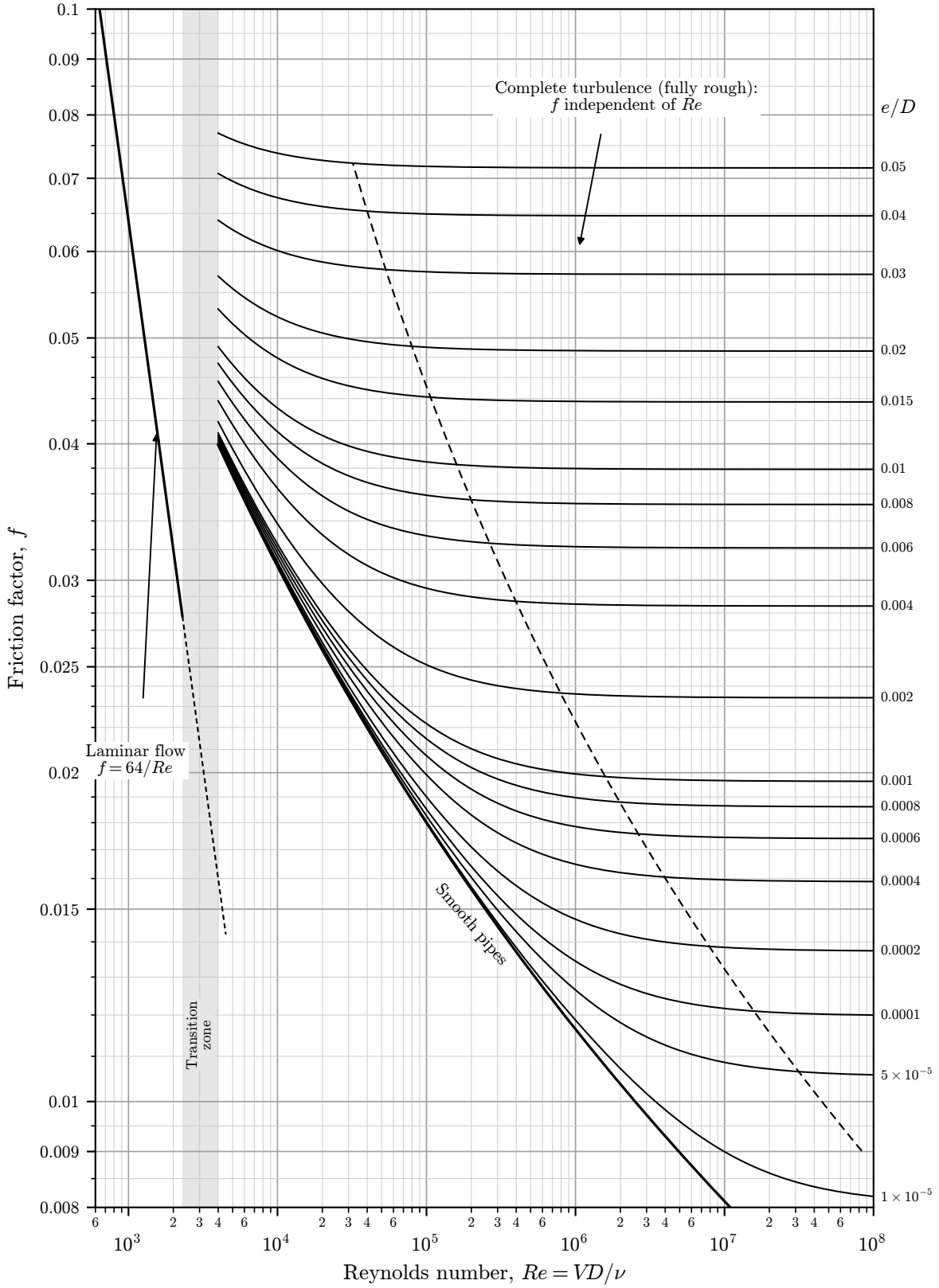


Figure 7: Moody chart: Darcy friction factor f computed from the Colebrook formula, with the laminar line $f = 64/Re$. Curves are labelled with the relative roughness e/D; to the right of the dashed line f is within 1% of its fully rough value.